Deploying a VM using an ARM Template in Azure SOP

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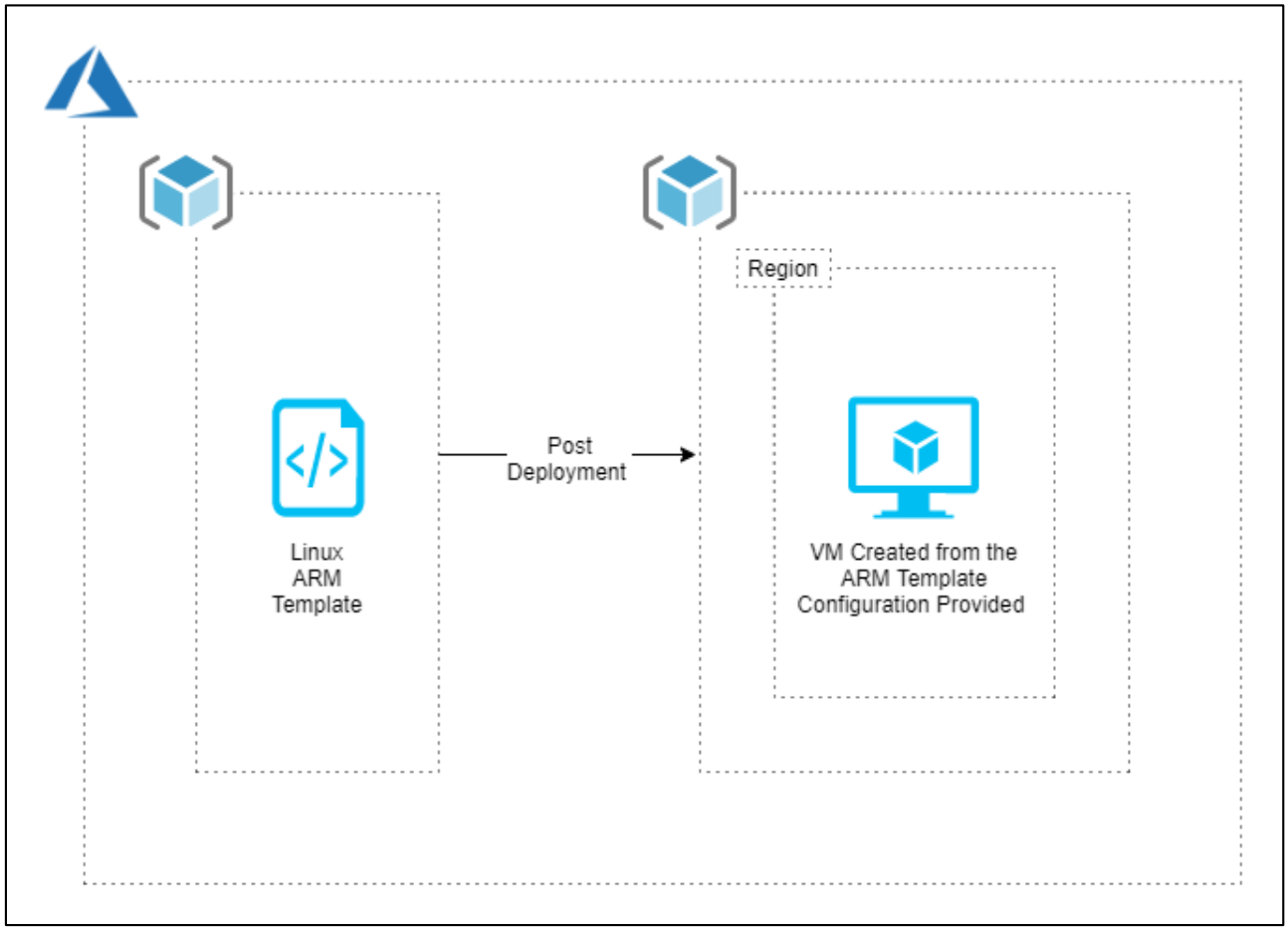
1. Deploying a VM using an ARM Template in Azure

1.1 Description

The objective of this lab exercise is to create and deploy a Virtual Machine (Linux) using the Azure Resource Manager (ARM). ARM Templates are a way to declare the objects, the types, names and properties in a JSON file which can be checked into source control and managed like any other code file. ARM Templates are what really gives the ability to roll out **Azure Infrastructure as code** which is the process of managing and provisioning computer data centres through machine-readable definition files.

1.2 Architecture Diagram

The diagram below displays a visual representation of the application architecture:



1.3 Lab Steps

Follow the steps outlined below to achieve the objective of this lab exercise:

1. Go to the **Resource group** for your user (<**Dept**><**empID**>**RG**).
2. Click **Add** and Search for **templates**. Choose **Template deployment (deploy using custom templates).**
3. Click **Create.**
4. Under Common templates, choose **Create a Linux virtual machine**.
5. If you are not able to select the above template, copy the json from the GitHub link displayed in message and paste it in **Build your Own template in the Editor**.
6. Under **Basics**, Select the Resource group as **AzureTeam1-RG** and Location as **East US**.
7. Under **Settings**, give the VM a name, Admin Username, choose authentication type as **password**, give a password for the same and set the VM Size as **Standard_B1s** according to the policy set.
8. Select the **I agree to the terms and conditions stated above** check box.
9. Select **Purchase.** Your VM will be created in a few minutes.

1.4 Troubleshooting

S. No	Problem	Solution
1	Not able to create the virtual machine due to authorization failure (Code: **AuthorizationFailed**)	Check beforehand the authorization given, to create this VM, if not take help from the team to give the required authorization.
2	Unable to create the virtual machine due to policy violations (Code: **RequestDisallowedByPolicy**)	Check the policy and give the valid details under the Basics and Settings that are allowed by policy.

1.5 Supporting References

Refer the below links for additional information:

1. https://docs.microsoft.com/en-us/azure/azure-resource-manager/templates/deploy-portal
2. https://docs.microsoft.com/en-us/azure/virtual-machines/windows/ps-template